Lesson 3: Take Your Shot

GDevelop

VIDEO GAME CREATION

This week: spawning objects at runtime. Your job is to make your player shoot a projectile that flies up the screen and cleans itself up when it leaves - the first piece of your capstone's actual shooting mechanic.

Open your own [YourName]Game project from Lesson 2 - don't start a new one. Everything this week builds on the player object you already have.

Tip: this lesson's Assets folder has a ready-to-use placeholder projectile image (bullet_yellow.png) and a new sound effect (sfx_shoot.wav) - you don't need to find or make your own art today.

Setup

1. Open your [YourName]Game project from Lesson 2.
2. Add a new Sprite object, name it Bullet, and assign bullet_yellow.png (import it from this lesson's Assets folder) as its image.
3. Select Bullet, click "Add a behavior," and add Destroy outside screen (default settings) - this is a built-in GDevelop behavior, already installed in every project.

Level 1 - Minimal (MVP)

Pressing the fire key spawns one projectile that flies upward and disappears when it leaves the screen. This is a complete finish for today - reaching this is enough to move on to next week.

4. Add a new event: Condition "Key pressed [Space]" -> Action "Create an object" (choose Bullet, positioned at Player.X(), Player.Y()).
5. Add a second event that always runs (no condition needed): Action "Change the object's Y position" for Bullet, subtracting a fixed amount times TimeDelta() each frame so it flies upward.
6. Click the green Preview button and press Space to test - a bullet should spawn at the player and fly upward, then disappear near the top on its own (the Destroy outside screen behavior handles the cleanup automatically, no event needed).

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Condition: Key pressed [Space]
Action: Create object Bullet at (Player.X(), Player.Y())

(always, no condition)
Action: Bullet.Y = Bullet.Y() - 400 * TimeDelta()
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(Bullet object has the Destroy outside screen behavior - handles cleanup
with zero events)
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Level 2 - Complete (Alpha/Beta)

Firing repeatedly produces multiple independent projectiles on screen at once, each moving and disappearing correctly, and rapid-fire is capped so holding the key down doesn't flood the screen.

- Test firing several times quickly - each press should produce its own independent Bullet instance, all moving and being destroyed correctly. Since "Create an object" makes a fully separate instance each time, and the movement/cleanup events apply to every Bullet instance automatically, this already works with no extra code.
- Add a cooldown so holding Space doesn't fire every frame: GDevelop's "Key pressed" condition is actually true on every frame the key is held (unlike some other engines' "pressed" checks), so without a cooldown this WILL flood the screen - add a scene Timer ("Start (or reset) a scene timer" once at the beginning of the scene), and change the fire condition to also require Timer value "FireCooldown" >= 0.2 seconds, resetting the timer on each shot.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: add a "Play a sound" action (using the imported sfx_shoot.wav) to the same event that creates a Bullet.
- Visual: give Bullet its own "Tint color" or a brief scale pulse via "Change the scale of an object" that fades back to normal after a fraction of a second, so it reads clearly as "a shot" rather than a copy of the player - not a permanent recolor.
- Structure: keep every Bullet-specific event (its movement, and anything from the ideas above) grouped together in one labeled event group, separate from the Player's own movement events from Lesson 2.

If Something's Wrong

- Bullets fire dozens of times per press: this is the "Key pressed" condition being level-triggered (true the whole time a key is held) rather than edge-triggered - see the Level 2 cooldown fix, which is required here, not optional.
- Bullet spawns but doesn't move: confirm the "Change the object's Y position" event has no condition attached (or a condition that's always true) - it needs to run every frame, not just on the frame the key was pressed.
- Bullet never disappears: confirm the Destroy outside screen behavior is actually attached to the Bullet object (check its Behaviors tab), not just described in a comment.

Want More?

Student Resource List > GDevelop section, or gdevelop.io/tutorials